Chromium Browser Artifact Parser

Investigator's Guide to Reviewing Output

Version: 1.0
Tool Author: Jacques Boucher (jjrboucher@gmail.com)
Repository: https://github.com/jjrboucher/Chromium-Browser-Artifact-Parser

---

Table of Contents

1. Introduction
2. Understanding the Excel Output
3. Critical Data Retention Information
4. Worksheet Reference Guide
5. Understanding Gap Analysis Worksheets
6. The Preferences Worksheet: User Identity Evidence
7. Cross-Referencing Data Across Worksheets
8. Interpreting Missing Worksheets
9. Annex A: Investigative Tips and Techniques
10. Annex B: URL Pattern Reference for Common Services
11. Annex C: Timestamp Quick Reference

---

1. Introduction

Purpose of This Guide

This guide is designed to assist forensic investigators, incident responders, and security analysts in effectively reviewing and interpreting the Excel output generated by the Chromium Browser Artifact Parser. Understanding how to properly analyze this data is crucial for drawing accurate conclusions in investigations.

What the Parser Extracts

The Chromium Browser Artifact Parser extracts forensic artifacts from Chromium-based browsers (Google Chrome, Microsoft Edge, Opera, Brave, Vivaldi, etc.) and presents them in a structured Excel workbook. Each worksheet contains a specific category of browser data, complete with decoded timestamps and descriptive field values.

Key Investigation Principles

Before diving into the data, keep these principles in mind:

1. **Timestamps are in UTC** - All decoded timestamps are in Coordinated Universal Time. Convert to local time as needed for your investigation timeline.
2. **Absence of evidence is not evidence of absence** - Missing data could indicate deletion, data that was never created, or data that aged out of retention.
3. **Correlate across worksheets** - A single artifact often tells only part of the story. Cross-reference timestamps and URLs across multiple worksheets to build a complete picture.
4. **Consider data retention limits** - Chromium browsers automatically purge certain data. Understanding these limits is critical for proper interpretation.

---

2. Understanding the Excel Output

Workbook Structure

The Excel output typically contains the following worksheets (in order):

1. **Summary** - Overview of record counts for each artifact category
2. **Preferences** - User identity and browser configuration data
3. **Artifact Worksheets** - Individual sheets for each processed artifact type

The Summary Worksheet

The Summary worksheet should be your first stop. It provides a quick overview:

Worksheet Name	Record Count
History	15,432
Downloads	287
Cookies	3,891
Login Data	45
...	...

Investigative Value:

- Quickly identify which artifact categories have data
- A record count of 0 may indicate the user cleared that data type or the artifact doesn't exist
- Unusually low counts (compared to expected usage) may suggest data deletion

Column Naming Convention

Most worksheets follow a consistent column naming pattern:

- **Raw timestamp columns** - Named after the database field (e.g., `visit_time`, `date_created`)
- **Decoded timestamp columns** - Prefixed with "Decoded" (e.g., `Decoded visit_time (UTC)`)
- **Decoded value columns** - Prefixed with "Decoded" for enumerated values (e.g., `Decoded danger_type`)

---

3. Critical Data Retention Information

The 90-Day History Limitation

This is one of the most important facts for investigators to understand:

> **Chromium-based browsers automatically delete browsing history older than approximately 90 days.**

This means:

- The History worksheet will only contain visits from roughly the last 3 months
- You cannot conclude that a user never visited a site if it doesn't appear in history - they may have visited it more than 90 days ago
- Older history has been automatically purged by the browser, not necessarily by user action

Data That Can Persist Beyond 90 Days

While history is limited to 90 days, other worksheets may contain data far older:

Worksheet	Retention Period	Notes
History	~90 days	Automatically purged
Downloads	Indefinite	Until user clears or database limit reached
Login Data	Indefinite	Saved credentials persist until deleted
Autofill	Indefinite	Form data persists until deleted
Bookmarks	Indefinite	Persist until manually deleted
Cookies	Varies	Based on expiration or session
Addresses	Indefinite	Until manually deleted
Credit Cards	Indefinite	Until manually deleted
Bank Accounts	Indefinite	Until manually deleted
Loyalty Cards	Indefinite	Until manually deleted
WebAssist (Edge)	**Potentially years**	See special note below

Special Case: Microsoft Edge WebAssist

The WebAssist worksheet (Edge only) can contain browsing data **far older than 90 days** - potentially spanning years of activity. This is because:

- WebAssist is a separate database from the main History database
- It is used by Edge's AI assistant features and web assist functionality
- It maintains its own navigation history independent of the standard history retention policy
- Microsoft designed this for feature functionality, not forensic purposes, but it creates a valuable artifact

Investigative Significance: If you are analyzing a Microsoft Edge profile, the WebAssist worksheet may be your only source of browsing activity from more than 90 days ago. Always check this worksheet when investigating Edge profiles for historical activity.

---

4. Worksheet Reference Guide

4.1 History Worksheet

Source Database: History

Contains: Complete browsing history with navigation details

Key Columns:

Column	Description	Investigative Value
urls.id	Unique URL identifier	Links to other tables
visits.id	Unique visit identifier	NULL if no corresponding visit
url	The visited URL	Primary evidence of site access
title	Page title	Context for what user viewed
visit_count	Total visits to this URL	Frequency of access
typed_count	Times URL was manually typed	Indicates intentional navigation
Visit Source	Local, Synced, Extension, Imported	Origin of the record
Transition Type	How user navigated to page	User action context
Decoded visit_time (UTC)	When visit occurred	Timeline evidence
visit_duration	Time spent on page (seconds)	User engagement level

Understanding Visit Source:

- **Local** - Activity originated on this device
- **Synced** - Activity synced from another device via Google account
- **Extension** - Navigation triggered by a browser extension
- **Firefox/IE/Safari Imported** - History imported from another browser

Understanding Transition Types:

- **Clicked on a link** - User followed a hyperlink
- **Typed URL** - User manually entered the URL
- **Clicked on suggestion** - User selected from omnibox suggestions
- **User submitted form data** - Navigation resulted from form submission
- **User reloaded page** - Page was refreshed

4.2 Downloads Worksheet

Source Database: History
Contains: Downloaded file information

Key Columns:

Column	Description	Investigative Value
target_path	Where file was saved	Local file location
current_path	Current file location	May differ if moved
referrer	Page that initiated download	Context of download
tab_url	URL of tab when download started	What user was viewing
Decoded start_time (UTC)	When download began	Timeline
Decoded end_time (UTC)	When download completed	Duration calculation
received_bytes / total_bytes	Download progress	Completion status
Decoded danger_type	Security assessment	Malware indicators
Decoded state	Complete or Interrupted	Success status
Download Source	Actual file URL	True origin of file

Danger Type Values:

- **Not Dangerous** - No security concerns
- **Dangerous File/URL/Content** - Security warnings present
- **User Validated** - User overrode security warning
- **Blocked** variants - Download was prevented

4.3 Search Terms Worksheet

Source Database: History + Web Data
Contains: Search queries performed by the user

Key Columns:

Column	Description	Investigative Value
`url`	URL containing the search	Search engine/site used
`keyword`	Search engine identifier	Where search was performed
`search term`	The actual search query	**User intent evidence**
`Decoded last_visit_time (UTC)`	When search occurred	Timeline

Important Limitations:

> **The Search Terms worksheet may not capture ALL searches.** It primarily captures searches performed through recognized search engines (Google, Bing, etc.) that Chrome integrates with the omnibox. Searches performed through on-site search boxes (e.g., searching within Amazon, eBay, or other websites) may not appear in this worksheet.

To find site-specific searches, examine the History worksheet and look for URLs containing query parameters (e.g., `?q=`, `?search=`, `?query=`).

4.4 Bookmarks Worksheet

Source Database: Bookmarks (JSON) + Bookmarks.bak (JSON)
Contains: User's saved bookmarks

Special Feature: This worksheet combines data from two sources:

- **Bookmarks** - Current bookmarks file
- **Bookmarks.bak** - Backup/previous version of bookmarks

The `Source` column indicates which file the bookmark came from.

Key Columns:

Column	Description	Investigative Value
Source	Bookmarks or Bookmarks.bak	Origin file
Type	folder or url	Entry type
Folder Path	Location in bookmark hierarchy	Organization
Name	Bookmark name	User-assigned label
URL	Bookmarked address	Sites of interest
Converted Date added (UTC)	When bookmark was created	Timeline
Converted Date last used (UTC)	When bookmark was last accessed	Recent activity

Investigative Value of Bookmarks.bak: The backup file may contain bookmarks that were subsequently deleted from the main file. Comparing entries between sources can reveal:

- Recently deleted bookmarks
- Changes to bookmark organization
- Historical interests that user may have tried to hide

4.5 Login Data Worksheet

Source Database: Login Data
Contains: Saved website credentials

Key Columns:

Column	Description	Investigative Value
signon_realm	Domain for the credentials	Sites with saved logins
origin_url	Specific login page URL	Login page location
username_value	Saved username	**User identity evidence**
Decoded date_created (UTC)	When credential was saved	Account creation/first login
Decoded date_last_used (UTC)	When credential was last used	Recent account access
times_used	Number of times used	Frequency of access
Decoded blacklisted_by_user	TRUE = "never save" selected	User chose not to save

Note: Password values are encrypted and cannot be decrypted by this tool.

4.6 Cookies Worksheet

Source Database: Network/Cookies
Contains: Browser cookies

Key Columns:

Column	Description	Investigative Value
name	Cookie name	Identifies cookie purpose
host_key	Domain that set cookie	Site interaction evidence
Decoded creation_utc (UTC)	When cookie was created	First site interaction
Decoded last_access_utc (UTC)	When cookie was last accessed	Recent site interaction
Decoded expires_utc (UTC)	When cookie expires	Session vs persistent

4.7 Autofill Worksheet

Source Database: Web Data
Contains: Form field auto-complete data

Key Columns:

Column	Description	Investigative Value
Name	Form field name	Type of information entered
User Input	Value user entered	**Direct user input evidence**
# of times used	Usage frequency	How often value was entered
Decoded date_created (UTC)	First time value was entered	Timeline
Decoded date_last_used (UTC)	Most recent use	Recent activity

Common Field Names and Their Significance:

- email - Email addresses used
- username - Usernames entered
- name, first-name, last-name - Identity information
- address, city, zip - Location information
- phone - Phone numbers
- company, organization - Employer/affiliation

4.8 Addresses Worksheet

Source Database: Web Data
Contains: Saved address book entries

Contains full address records including name, email, phone, street address, city, state, postal code, and country. This is particularly valuable for identifying the user and their known locations.

4.9 Credit Cards Worksheet

Source Database: Web Data
Contains: Masked credit card information

Only the last four digits of card numbers are stored. This worksheet reveals:

- Financial institutions the user has accounts with
- Card networks (Visa, Mastercard, etc.)
- Expiration dates
- Account nicknames that may provide context
- Usage count and last use date

4.10 Bank Accounts Worksheet

Source Database: Web Data
Contains: Masked bank account information with usage metadata

Key Columns:

Column	Description	Investigative Value
instrument_id	Unique identifier	Links related records
Bank Name	Financial institution	Identifies bank relationship
Account Number Suffix	Last digits of account	Partial account identification
Account Type	Checking, savings, etc.	Account category
nickname	User-assigned name	Personal context
use_count	Number of times used	Frequency of use
Decoded use_date (UTC)	Last time account was used	Recent activity

Investigative Significance:

- Reveals which financial institutions the user has relationships with
- Usage data shows when and how often accounts were used for autofill
- Nicknames may reveal personal information or account purposes

4.11 Loyalty Cards Worksheet

Source Database: Web Data
Contains: Store loyalty and rewards card information

Key Columns:

Column	Description	Investigative Value
merchant_name	Store/retailer name	Shopping patterns
merchant_domain	Store website	Online shopping activity
program_name	Loyalty program name	Membership identification
program_logo	Logo URL	Visual identification
loyalty_card_number	Card/membership number	**Direct identifier**

Investigative Significance:

- Reveals stores where the user has loyalty memberships
- Loyalty card numbers can be used to obtain purchase history from retailers (with proper legal process)
- Merchant domains indicate online shopping patterns
- Can establish geographic patterns based on regional retailers

4.13 Shortcuts Worksheet

Source Database: Shortcuts
Contains: Omnibox (address bar) shortcuts

Key Columns:

Column	Description	Investigative Value
text	What user typed	Search/navigation behavior
url	Resulting URL	Destination
number_of_hits	Times this shortcut was used	Frequency
Decoded last_access_time (UTC)	Most recent use	Recent activity
Transition Type	How navigation occurred	User action

4.14 Top Sites Worksheet

Source Database: Top Sites
Contains: Most frequently visited sites

Shows sites that appear on Chrome's new tab page. Useful for quickly identifying the user's most visited destinations.

4.15 FavIcons Worksheet

Source Database: Favicons
Contains: Website icons and associated URLs

Investigative Value:

- page URL column reveals sites visited (even if history was cleared, favicons may persist)
- Decoded last_updated (UTC) indicates when the favicon was last refreshed
- Favicon data can persist after history is cleared

4.16 Keywords Worksheet

Source Database: Web Data
Contains: Search engine definitions

Shows configured search engines. The Decoded date_created (UTC) indicates when a search engine was first used, which can reveal when a user first performed a search on a particular site with a search function.

4.17 WebAssist Worksheet (Edge Only)

Source Database: WebAssistDatabase
Contains: Edge-specific navigation history

Key Columns:

Column	Description	Investigative Value
url	Visited URL	Site access evidence
title	Page title	Content context
decoded last_visited_time (UTC)	Visit timestamp	Timeline
num_visits	Visit count	Frequency

Critical Note: As discussed in Section 3, this worksheet can contain data **far older than the 90-day history limit**. For Edge investigations, this may be your primary source of historical browsing evidence.

---

5. Understanding Gap Analysis Worksheets

What Are Gap Worksheets?

The parser includes three "Gaps" worksheets:

- **History Gaps**
- **Downloads Gaps**
- **Login Data Gaps**

These worksheets identify **missing records** in the database by detecting gaps in the sequential record numbering (ROWID). SQLite databases assign incrementing IDs to records, so gaps in the sequence indicate that records once existed but have been deleted.

How to Read Gap Data

Each gap record contains:

Column	Description
Previous Record Number	The ID of the record that exists BEFORE the gap
Record Number	The ID of the record that exists AFTER the gap
Number of Missing Records	How many records are missing in this gap
Beginning Timestamp (UTC)	Timestamp of the record BEFORE the gap
Ending Timestamp (UTC)	Timestamp of the record AFTER the gap

Interpreting Gap Data

Example Gap Record:

Previous Record Number	Record Number	Number of Missing Records	Beginning Timestamp	Ending Timestamp
1523	1530	6	2024-03-15 14:32:00	2024-03-15 16:45:00

Interpretation:

- Records 1524, 1525, 1526, 1527, 1528, and 1529 are missing

- The existing record #1523 has a timestamp of 2024-03-15 14:32:00
- The existing record #1530 has a timestamp of 2024-03-15 16:45:00
- **Conclusion:** The 6 missing records represent activity that occurred **between** 14:32:00 and 16:45:00 on March 15, 2024

What Gaps Can Tell You

1. **Evidence of Deletion** - Gaps indicate records were deleted, either by:
    - User manually clearing browsing data
    - User using private/incognito mode (though this typically doesn't create gaps)
    - Third-party cleaning software
    - Targeted deletion of specific entries
2. **Timeframe of Deleted Activity** - The beginning and ending timestamps bracket when the deleted activity occurred
3. **Volume of Deleted Data** - The number of missing records indicates how much data was deleted in that gap
4. **Pattern Analysis** - Multiple gaps with similar timestamps may indicate periodic clearing

Special Cases in Gap Analysis

Gaps at the Beginning: If the first record in the database is not #1, the gap analysis will show missing records from the beginning. This indicates the earliest records have been deleted.

Gaps at the End: The parser checks if the highest existing record number is lower than the highest-ever-assigned number (stored in `sqlite_sequence`). If so, this indicates records at the end have been deleted.

No Gaps Found: An empty Gaps worksheet means the record numbering is sequential with no missing records. However, this doesn't necessarily mean no deletion occurred - the user could have cleared ALL history, resulting in a fresh database with sequential numbering starting over.

Limitations of Gap Analysis

- Gap analysis cannot tell you **what** was deleted, only **when** and **how many**
- If a user clears all browsing data, the database is reset and gaps from that clearing will not be detected
- Some gaps may result from browser crashes or corruption, not intentional deletion
- The Downloads database does not track its sequence number in `sqlite_sequence`, so gaps at the end cannot be detected for downloads

---

6. The Preferences Worksheet: User Identity Evidence

Overview

The Preferences worksheet extracts data from the browser's Preferences JSON file. This is one of the most valuable worksheets for **identifying the user** associated with a browser profile.

Key Evidence Types

User Identity Information

If the user was logged into their Google account (Chrome) or Microsoft account (Edge), you will find:

Field	Description	Example
Email	User's account email	john.doe@gmail.com
Full name	Account display name	John Doe
Given name	First name	John
gaia	Google Account ID (unique identifier)	123456789012345678901
Thumbnail URL	Profile picture URL	https://lh3.google...

Investigative Significance:

- **If the user is logged in, you can definitively identify whose browser profile this is**
- The email address links this profile to a specific individual
- The GAIA ID is a unique identifier that persists across sessions

Edge-Specific Identity Fields

For Microsoft Edge profiles:

- Edge First Name
- Edge Last Name

These fields are extracted from the Microsoft account associated with the Edge profile.

Profile Information

Field	Description
Profile name	Name of the browser profile
Profile created	When this profile was first created
Profile created using browser version	Browser version at profile creation
Previous Navigation Time	Last navigation before browser close

User Behavior Settings

Field	Description	Investigative Value
Default Download Directory	Where files are saved	File location evidence
Prompt for Download	Whether user is asked for location	Download behavior
Last "Save As" Directory	Most recent save location	Recent file activity
Startup value	What happens when browser opens	Usage patterns
URL startup list	Sites opened automatically	Regular destinations
Homepage	Home button URL	Preferred starting point
New tab shortcuts	Custom new tab links	Sites of interest

Exit Status

Field	Value	Meaning
Exit Type	Normal	Browser was closed properly
Exit Type	Crashed	Browser terminated unexpectedly

A "Crashed" exit type may indicate:

- Browser crashed during use
- System crashed while browser was open
- Browser process was terminated (e.g., by task manager)
- Potentially suspicious termination to hide activity

Country Information

The `Country ID at install` field indicates the country setting when the browser was installed. This can help establish:

- Geographic location of installation
- Language/locale settings
- Potential timezone context

---

7. Cross-Referencing Data Across Worksheets

The Power of Timestamp Correlation

One of the most effective investigation techniques is correlating timestamps across multiple worksheets to build a complete picture of user activity.

Technique: Activity Context from History

When you find an interesting artifact in any worksheet with a timestamp, navigate to the **History worksheet** and filter or search for activity around that same time. This reveals what the user was browsing that led to or coincided with the artifact.

Example Scenario: Investigating a Downloaded File

1. In the **Downloads** worksheet, you find:
    - File: `suspicious_tool.exe`
    - Start time: `2024-03-15 14:35:22 UTC`
    - Referrer: `https://example-forum.com/thread/12345`
2. Go to the **History** worksheet and filter for activity between `14:30:00` and `14:40:00` on `2024-03-15`
3. You discover:
    - 14:32:15 - User visited `https://example-forum.com/`
    - 14:33:45 - User viewed `https://example-forum.com/thread/12345` (the referrer!)
    - 14:35:22 - Download initiated
    - 14:36:00 - User visited `https://virus-scan.com` (checking the file?)

Now you have context: The user was browsing a forum, found a thread, downloaded a file from it, and then visited a virus scanning site.

Technique: Identifying Related Accounts

Cross-reference these worksheets to identify accounts associated with the user:

1. **Login Data** - Shows sites where user has saved credentials
2. **Autofill** - Shows email addresses and usernames entered in forms
3. **Addresses** - Shows personal information
4. **Preferences** - Shows logged-in account
5. **Cookies** - Shows sites with persistent sessions

Technique: Verifying Claimed Activity

If a user claims they visited a certain site at a certain time:

1. Check **History** for the visit
2. Check **Cookies** for session evidence
3. Check **FavIcons** for site access evidence
4. Check **Shortcuts** for address bar activity
5. Check **Gap Analysis** to see if relevant timeframes show deletion

Technique: Building User Profiles from Searches

Combine search-related worksheets:

1. **Search Terms** - Direct search queries
2. **Keywords** - Search engines used (indicates which sites have search boxes the user has used)
3. **History** - URLs with query parameters
4. **Autofill** - Search terms entered in form fields

---

8. Interpreting Missing Worksheets

Why a Worksheet Might Be Missing

A worksheet will not appear in the output if:

1. **Zero records exist** - The database table is empty
2. **Database file doesn't exist** - The expected file isn't in the profile
3. **Processing error** - An error occurred (check the Summary for "0" counts)
4. **Not selected for processing** - The artifact wasn't checked in the parser

Investigative Interpretations

Missing Worksheet	Possible Explanations
History	User cleared all history; New profile; History disabled
Downloads	User never downloaded files; User cleared downloads; Files deleted
Login Data	User doesn't save passwords; User cleared saved passwords
Cookies	User cleared cookies; Cookies disabled; Privacy mode used
Search Terms	No recognized searches performed; Search history cleared
Autofill	Autofill disabled; User cleared autofill data
Bookmarks	User has no bookmarks
Credit Cards / Bank Accounts	User hasn't saved financial information
WebAssist	Not an Edge profile; Edge feature not used

The Significance of Zero Records vs. Missing Data

Check the **Summary** worksheet:

- **Worksheet listed with 0 records** - The database exists but contains no data. This could indicate:
  - Intentional clearing of that specific data type
  - Feature was enabled but never used
- **Worksheet not listed at all** - The artifact wasn't processed or the database file doesn't exist

Corroborating Evidence of Deletion

If you suspect a user cleared data:

1. **Check Gap Analysis** - Are there gaps in remaining data?
2. **Check Preferences** - Look at `privacy_settings` / `clear_data` fields
3. **Compare artifact dates** - Do all artifacts suddenly start from the same date?
4. **Check other worksheets** - Did clearing miss anything (FavIcons often survive)?

Annex A: Investigative Tips and Techniques

A.1 Essential Excel Skills for Browser Forensics

Filtering:

- Use Excel's AutoFilter (Ctrl+Shift+L) on each worksheet
- Filter columns to isolate specific URLs, domains, or timeframes
- Use "Contains" filters for partial URL matching

Sorting:

- Sort by timestamp columns to create chronological views
- Sort by `visit_count` or `number_of_hits` to find frequently accessed content

Conditional Formatting:

- Highlight rows with specific keywords
- Color-code timestamps within investigation timeframes
- Highlight "dangerous" download classifications

Find and Replace (Ctrl+H):

- Quickly locate specific domains or terms
- Use wildcards for pattern matching

A.2 URL Filtering Strategies

Identifying Specific Services:

Use the filter function on URL columns with "Contains" to find:

To Find	Filter For	Column to Filter
Gmail inbox access	mail.google.com/mail containing inbox	url
Google Drive activity	drive.google.com	url
Google Docs	docs.google.com/document	url
Google Sheets	docs.google.com/spreadsheets	url
YouTube viewing	youtube.com/watch	url
Facebook	facebook.com	url
Twitter/X	twitter.com or x.com	url
LinkedIn	linkedin.com	url
Outlook web	outlook.live.com or outlook.office.com	url
OneDrive	onedrive.live.com	url
Dropbox	dropbox.com	url
Amazon	amazon.com	url
Banking (general)	bank or common bank names	url

Identifying File Downloads by Type:

Filter the target_path or Download Source column in Downloads:

File Type	Filter Pattern
Executables	.exe
Archives	.zip, .rar, .7z
Documents	.pdf, .docx, .xlsx
Images	.jpg, .png, .gif
Torrents	.torrent

A.3 Timeline Analysis Techniques

Creating a Master Timeline:

1. Copy timestamp and key columns from multiple worksheets into a new sheet
2. Add a "Source" column to identify origin worksheet
3. Sort all entries by timestamp
4. Review chronologically to see activity patterns

Identifying Activity Clusters:

Look for periods of intense activity followed by gaps:

- Multiple downloads in quick succession may indicate bulk downloading
- Dense browsing followed by long gaps may indicate work sessions
- Activity outside normal hours may be significant

A.4 Identifying Suspicious Activity

Indicators of Potential Data Exfiltration:

- Downloads of archive files (.zip, .rar)
- Access to cloud storage services
- Uploads to file sharing sites
- Email attachment activity
- USB/external storage paths in download locations

Indicators of Policy Violations:

- Access to unauthorized sites
- Downloads of prohibited content
- Personal activity during work hours
- Use of anonymization services (VPN sites, Tor references)

Indicators of Cover-Up Attempts:

- Large gaps in History Gaps worksheet
- Multiple gaps clustered at similar times
- Clearing data before specific events
- "Crashed" exit status at unusual times

A.5 Validating Findings

Corroborate with Multiple Sources:

- Don't rely on a single worksheet for conclusions
- Cross-reference timestamps across artifacts
- Check if Cookies support claimed History visits
- Verify downloads match referrer URLs

Document Negative Findings:

- Note when expected evidence is absent
- Record which artifacts have gaps
- Document what was searched for but not found

A.6 Search Query Analysis

Extracting Intent from Search Terms:

The Search Terms worksheet reveals user intent. Look for:

Search Type	Investigative Significance
Technical searches	Skills, interests, projects
Location searches	Places of interest, travel plans
Person searches	Relationships, research on individuals
Legal/medical searches	Personal situations, concerns
Job searches	Employment status, dissatisfaction
Product searches	Purchases, interests

When Search Terms Are Incomplete:

Remember that the Search Terms worksheet primarily captures searches through integrated search engines. For site-specific searches:

1. Go to History worksheet
2. Filter for the site in question (e.g., `amazon.com`)
3. Look for URLs containing `search`, `query`, `q=`, or `?s=`

4. The query parameter typically contains the search term

Example: An Amazon search URL might look like:

```
https://www.amazon.com/s?k=wireless+keyboard
```

The search term is `wireless keyboard` (the `k=` parameter).

A.7 Leveraging Financial and Loyalty Data

Bank Accounts and Credit Cards:

The Bank Accounts and Credit Cards worksheets reveal financial relationships:

- Cross-reference bank names with History to find online banking activity
- Use account nicknames for context (e.g., "Joint Checking", "Business Card")
- The `use_count` and `use_date` fields show how often accounts were used for autofill

Loyalty Cards - A Hidden Gem:

The Loyalty Cards worksheet can be particularly valuable:

1. **Identifying Shopping Patterns:**
   - Merchant names reveal where the user shops
   - Multiple cards from similar retailers may indicate preferences
2. **Obtaining Purchase History:**
   - Loyalty card numbers can be used (with proper legal process) to request purchase history from retailers
   - This may reveal purchases, locations visited, and timestamps
3. **Geographic Profiling:**
   - Regional retailers (grocery stores, pharmacies) can indicate where the user lives or frequently visits
   - National chain loyalty cards may show travel patterns
4. **Correlating with Other Evidence:**
   - Match merchant domains with History worksheet to see when user visited those stores online
   - Cross-reference with Downloads for receipts or order confirmations

---

Annex B: URL Pattern Reference for Common Services

Email Services

Service	URL Pattern	Indicates
Gmail Inbox	mail.google.com/mail/u/0/#inbox	Viewing inbox
Gmail Sent	mail.google.com/mail/u/0/#sent	Viewing sent mail
Gmail Compose	mail.google.com/mail/u/0/?compose=	Composing email
Gmail Search	mail.google.com/mail/u/0/#search/	Searching email
Outlook Inbox	outlook.live.com/mail/0/inbox	Viewing inbox
Outlook Sent	outlook.live.com/mail/0/sentitems	Viewing sent
Yahoo Mail	mail.yahoo.com/d/folders/1	Viewing inbox

Cloud Storage

Service	URL Pattern	Indicates
Google Drive Home	drive.google.com/drive/	Browsing files
Google Drive File	drive.google.com/file/d/	Viewing specific file
OneDrive	onedrive.live.com/?id=	Viewing files
Dropbox	dropbox.com/home	Viewing files
Dropbox Shared	dropbox.com/sh/	Accessing shared folder
Box	app.box.com/	Viewing files
iCloud	icloud.com/iclouddrive	Viewing files

Social Media

Service	URL Pattern	Indicates
Facebook Profile	facebook.com/profile	Viewing own profile

Service	URL Pattern	Indicates
Facebook User	facebook.com/[username]	Viewing another user
Facebook Messenger	facebook.com/messages	Messaging
Twitter/X Timeline	twitter.com/home or x.com/home	Viewing feed
Twitter/X Profile	twitter.com/[username]	Viewing profile
LinkedIn Profile	linkedin.com/in/	Viewing profile
LinkedIn Messages	linkedin.com/messaging	Messaging
Instagram	instagram.com/	Viewing content
TikTok	tiktok.com/	Viewing content

Productivity/Documents

Service	URL Pattern	Indicates
Google Docs	docs.google.com/document/d/	Editing document
Google Sheets	docs.google.com/spreadsheets/d/	Editing spreadsheet
Google Slides	docs.google.com/presentation/d/	Editing presentation
Microsoft Office Online	office.com/launch/	Opening Office app
Word Online	word.office.com	Editing Word doc
Excel Online	excel.office.com	Editing Excel

File Sharing / Transfer

Service	URL Pattern	Indicates
WeTransfer	wetransfer.com	File transfer
SendGB	sendgb.com	File transfer
MediaFire	mediafire.com/file/	File access
Mega	mega.nz/	File access

Service	URL Pattern	Indicates
Google Drive Share	drive.google.com/file/d/[id]/view	Shared file

Annex C: Timestamp Quick Reference

Understanding Timestamp Columns

All "Decoded" timestamp columns are in **UTC (Coordinated Universal Time)**.

To convert to local time:

1. Note the UTC timestamp
2. Apply the appropriate timezone offset
3. Account for Daylight Saving Time if applicable

Example:

- Decoded timestamp: 2024-03-15 14:30:00 UTC
- Local timezone: Eastern (UTC-5) / Eastern Daylight (UTC-4)
- March 15 is during Daylight Saving Time
- Local time: 2024-03-15 10:30:00 EDT

Common Timestamp Fields by Worksheet

Worksheet	Primary Timestamp Field	What It Represents
History	Decoded visit_time	When page was visited
Downloads	Decoded start_time	When download began
Downloads	Decoded end_time	When download completed
Downloads	Decoded last_access_time	When downloaded file was opened
Cookies	Decoded creation_utc	When cookie was created
Cookies	Decoded last_access_utc	When cookie was last used
Login Data	Decoded date_created	When credential was saved
Login Data	Decoded date_last_used	When credential was last used

Worksheet	Primary Timestamp Field	What It Represents
Autofill	Decoded date_created	When value was first entered
Autofill	Decoded date_last_used	When value was last used
Bookmarks	Converted Date added	When bookmark was created
Bookmarks	Converted Date last used	When bookmark was accessed
FavIcons	Decoded last_updated	When favicon was refreshed
Shortcuts	Decoded last_access_time	When shortcut was used
Keywords	Decoded date_created	When search engine was first used
WebAssist	decoded last_visited_time	When page was visited

Interpreting Zero or Missing Timestamps

Value	Meaning
0 or empty in raw field	Never occurred / Not applicable
"Not opened via Chrome"	File was downloaded but not opened through Chrome
"Synced. Not used on this device"	Credential synced from another device, not used locally
"Never"	Action (like password modification) never occurred

Document Information

Document Title: Investigator's Guide to Reviewing Chromium Browser Artifact Parser Output
Version: 1.0
Last Updated: January 2026
Tool Repository: https://github.com/jjrboucher/Chromium-Browser-Artifact-Parser
Tool Author: Jacques Boucher (jjrboucher@gmail.com)

This guide is intended to assist trained forensic investigators. Always follow proper forensic procedures, maintain chain of custody documentation, and consult with legal counsel regarding admissibility of evidence in your jurisdiction.